

27/07/26

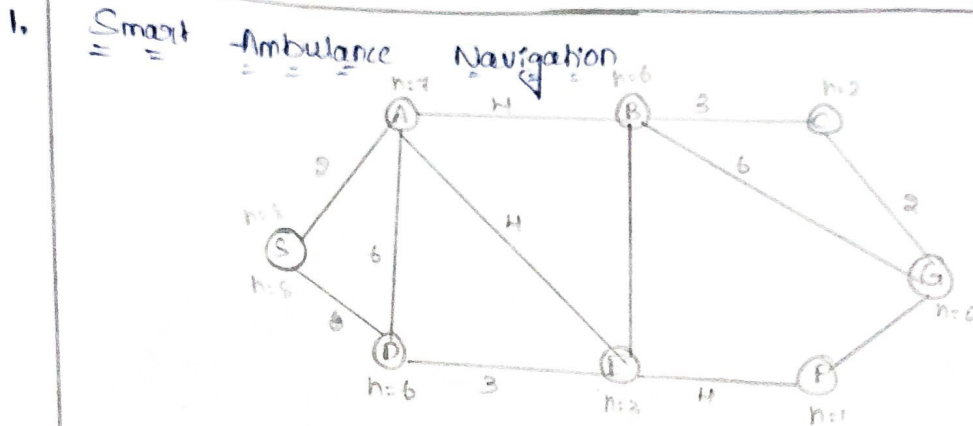
CO2 - A7-2

S. Venkatesh waslu

1920253410

K4A0101 - Artificial

Intelligence & expert system



Sol: Let us use the A* evaluation function

$$f(n) = g(n) + h(n)$$

* $h(n)$ → heuristic function

* $g(n)$ → actual cost

Step-1

$$S \rightarrow A = 0 + 8 = 8 \quad \checkmark$$

$$S \rightarrow D = 2 + 7 = 9$$

Step-2

$$S \rightarrow A \rightarrow B = 2 + 4 = 6 + 6 = 12 \quad \checkmark$$

$$S \rightarrow A \rightarrow D = 8 + 6 = 14$$

Step-3

$$g(E) = 2 + 4 = 6$$

$$h(E) = 3$$

$$f(E) = 6 + 3 = 9$$

Step-4

$$g(B) = 6 + 2 = 8$$

for D

$$g(D) = 6 + 3 = 9$$

$$g(D) = 8$$

10/8/26

for f

$$g(f) = 6 + 4 = 10$$

$$h(f) = 1$$

$$f(f) = 10 + 1 = 11$$

Step-5: Expand f

$$g(G) = 10 + 1 = 11$$

$$h(G) = 0$$

$$f(G) = 11 + 0 = 11$$

$S \rightarrow A \rightarrow E \rightarrow F \rightarrow G$

Total cost

$$2 + 4 + 4 + 1 = 11$$

Total route cost = 11

2.

Autonomous

Warehouse

Robot

5	S	?	?	?	?	?	?
4	?	#	?	?	#	?	?
3	?	?	?	#	?	?	#
2	?	#	?	?	?	#	?
1	?	?	#	?	?	?	G
	1	2	3	4	5	6	7

Given

* Start (S) = col 1, Row 5.

* Goal (G) = col 7, Row 1

The robot doesn't know all obstacles initially

* Use an A* Search

Search Strategy

$$f(n) = g(m) + h(n)$$

we use Manhattan distance

$$h(n) = |x_n - x_{goal}| + |y_n - y_{goal}|$$

Step-1 Initial decision

The robot starts at

$$S = (1, 5)$$

Goal

$$G = (7, 1)$$

$$S \rightarrow (2, 5) \rightarrow (3, 5) \rightarrow (4, 5) \rightarrow (5, 5) \rightarrow (6, 5) \rightarrow (7, 5)$$

Step-2 :- Obstacle discovery

When robot reaches $(5, 5)$, it observes the neighboring cell $(5, 4)$

It discovers that $(5, 4) = \#$

Robot's reaction

known obstacle : $(5, 4)$

Then a* is executed

new route : $(6, 5) \rightarrow (6, 4) \rightarrow (7, 4) \rightarrow (7, 3)$

Step-3 reaches $(7, 3) = \#$

The new route becomes : $(7, 4) \rightarrow (6, 4) \rightarrow (6, 3) \rightarrow (6, 2)$

Step-4

At $(6, 3)$ the robot discovers

$$(6, 2) = \#$$

It now finds

$$(6, 3) \rightarrow (5, 3) \rightarrow (5, 2) \rightarrow (5, 1)$$

then

$$(5, 1) \rightarrow (6, 1) \rightarrow (7, 1)$$

and reaches the goal

* final route followed by the route.

one valid online - search

$S \rightarrow (2,5) \rightarrow (3,5) \rightarrow (4,5) \rightarrow (5,5) \rightarrow (6,5) \rightarrow (7,5) \rightarrow (7,4) \rightarrow (6,4) \rightarrow (6,3) \rightarrow (5,2) \rightarrow (6,1) \rightarrow (6,1) \rightarrow G$

3.

University Examination Scheduling:

a) CSP formulation.

Variables:

$X = \{E_1, E_2, E_3, E_4, E_5, E_6\}$

Domains

$D(E_i) = \{T_1, T_2, T_3, T_4\}$

initially:

Exam

Domain

E_1

T_1, T_2, T_3, T_4

E_2

T_1, T_2, T_3, T_4

E_3

T_1, T_2, T_3, T_4

E_4

T_1, T_2, T_3, T_4

E_5

T_1, T_2, T_3, T_4

E_6

T_1, T_2, T_3, T_4

Constraints

$E_1 \neq E_2$

$E_1 \neq E_3$

$E_2 \neq E_4$

$E_3 \neq E_5$

$E_4 \neq E_6$

$E_5 \neq E_6$

Student Capacities

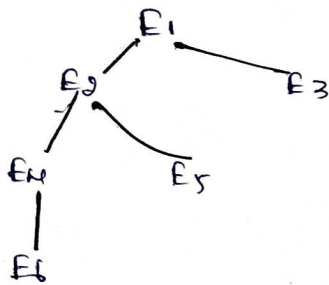
Exam Student's

E ₁	100
E ₂	80
E ₃	70
E ₄	60
E ₅	90
E ₆	50

Hall Capacity

Hall	Capacity
A	120
B	90
C	60

Conflict Graph



D) MRV + Forward Checking

Step-1

$$E_1 = T_1$$

$$D(E_2) = \{T_2, T_3, T_4\}$$

$$D(E_3) = \{T_2, T_3, T_4\}$$

Current domains

<u>Exam</u>	<u>Domain</u>
E ₁	T ₁
E ₂	T ₂ , T ₃ , T ₄
E ₃	T ₂ , T ₃ , T ₄
E ₄	T ₁ , T ₂ , T ₃ , T ₄
E ₅	T ₁ , T ₂ , T ₃ , T ₄
E ₆	T ₁ , T ₂ , T ₃ , T ₄

Step-2

$$C_2 = T_2$$

$$D(E_4) = \{T_1, T_2, T_4\}$$

$$D(E_1) = \{T_1, T_3, T_4\}$$

Step-3

$$C_3 = T_2$$

Therefore

$$D(E_5) = \{T_1, T_3, T_4\}$$

Step-4

$$\text{Assign: } C_4 = T_1$$

Therefore,

$$D(E_6) = \{T_2, T_3, T_4\}$$

Step-5

$$\text{Assign: } C_5 = T_1$$

Check

$$C_5 \neq C_2 \rightarrow T_1 \neq T_2$$

$$C_5 \neq C_3 \rightarrow T_1 \neq T_2$$

Step-6

$$\text{Assign } C_6 = T_2$$

Check

$$C_6 \neq C_4$$

$$T_2 \neq T_1$$

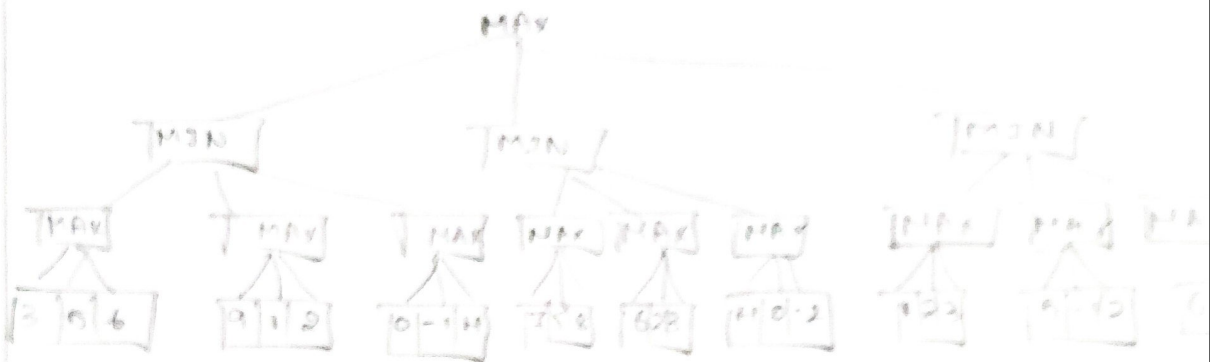
Conflict Verification

<u>constraint</u>	<u>Result</u>
$C_1 \neq C_2$	$T_1 \neq T_2$
$C_1 \neq C_3$	$T_1 \neq T_2$
$C_2 \neq C_4$	$T_2 \neq T_1$
$C_2 \neq C_5$	$T_2 \neq T_1$
$C_4 \neq C_6$	$T_1 \neq T_2$
$C_3 \neq C_5$	$T_2 \neq T_1$

$$\boxed{E_1 = T_1, E_2 = T_2, E_3 = T_2, E_4 = T_1, E_5 = T_1, E_6 = T_1}$$

Game - playing AI - Minimax & Alpha-Beta

H0



1. problem - Analysis

* MAX - AI

* MIN - opponent

MAX attempts to maximize the final score.

2. Minimax Rule

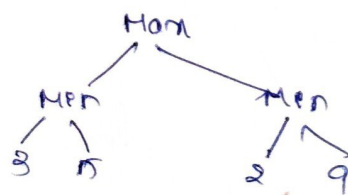
At Max:

$$V(n) = \max(v(\text{child}_1), v(\text{child}_2), \dots)$$

At Min:

$$V(n) = \min(v(\text{child}_1), v(\text{child}_2), \dots)$$

3. Example Tree



Left Min:

$$\min(3, 5) = 3$$

Right (Min)

$$\min(2, 9) = 2$$

Root (Max)

$$\max(3, 2) = 3$$

∴ Max selects the left branch

Alpha-Beta Analysis

Initially

$$\alpha = -\infty$$

$$\beta = +\infty$$

Suppose the first MIN branch gives MAX a value of 3

Therefore: $\boxed{\alpha = 3}$

When evaluating the next MIN node; if its first child gives 2

then

$$\beta = 2$$

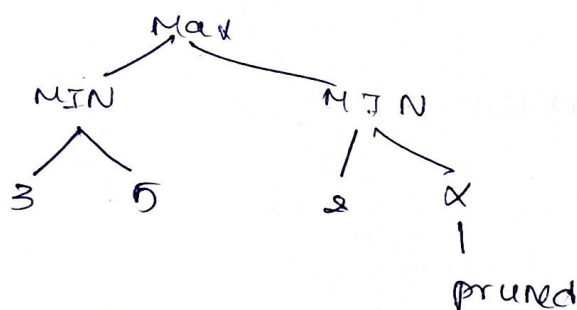
now

$$\alpha \geq \beta$$

because

$$3 \geq 2$$

Therefore, the remaining child of that MIN node can be pruned



The value of x doesn't affect the final decision

Delivery Route Optimization

5.

1. problem Analysis

min Distance (Route)

for minimization

Best Neighbours = arg min Distance (Neighbours)

2. Analytical Example

$D = 100 \text{ km}$

Route	Distance
R_1	95 km
R_2	10 km
R_3	90 km

$$90 < 95 < 100 < 100$$

$$\therefore 100 \rightarrow 90$$

Now Suppose D_3 's Neighbour

Route	Distance
R_4	85 km
R_5	93 km
R_6	88 km

select R_4

$$90 \rightarrow 85$$

Suppose R_{N4} neighbours are

<u>Route</u>	<u>Distance</u>
R ₁	80 km
R ₂	87 km
R ₉	84 km

select R₁

85 $\rightarrow$ 80

IF all neighbours of R₁ have $\geq$ to 80 km

Distance (R₁) = 80

then algorithm stops $\rightarrow$

